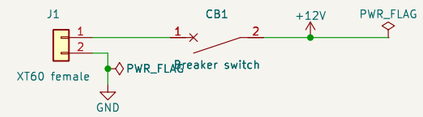
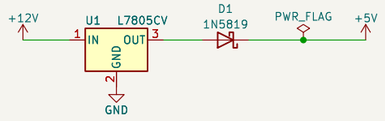
Power input and breaker switch



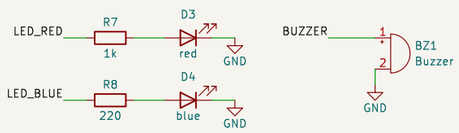
J1: female XT60 on the case, for the battery (LiFePO4 12.8 V 6 Ah with BMS) or a lab supply (12–14 V, current limit about 2 A).
The battery is charged off the mount.
CB1: resettable breaker switch, 3–5 A, rated for DC.

5 V supply for the ESP32 board



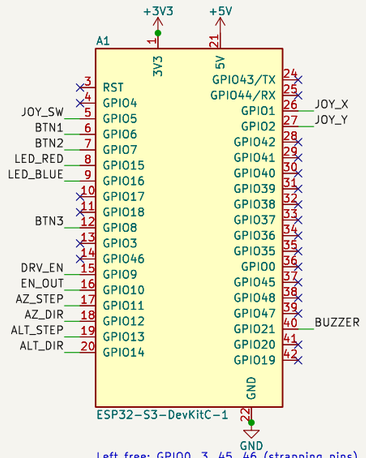
U1 on a small TO-220 heatsink (tab = GND), about 1 W.
Place it near C6/C7, which also serve as its input capacitors. D1 stops U1 and the board's USB 5 V from feeding each other.

LEDs and buzzer (on the case)



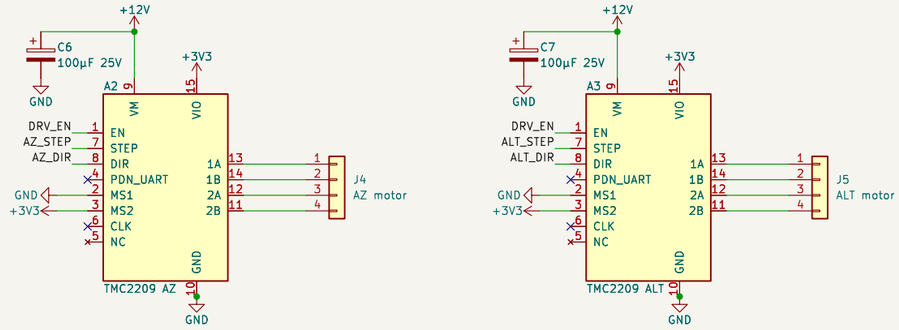
GPIO high = LED or buzzer on. PWM sets the LED brightness, or the tone of a passive buzzer.

Controller



Left free: GPIO0, 3, 45, 46 (strapping pins), GPIO19/20 (USB), GPIO35–37 (PSRAM on some boards), GPIO43/44 (USB serial console).

Stepper drivers



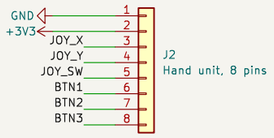
Standalone mode, no UART: MS1 = GND, MS2 = 3V3 gives 1/64 microsteps.
Motor current: potentiometer (Vref) on each module.
C6, C7 must sit close to the VM pins: they absorb the motor voltage spikes.

E-stop (on the case)



GPIO10 drives EN_OUT low to enable the drivers.
E-stop pressed or ESP32 in reset: R3 pulls DRV_EN high and both drivers switch off.
GPIO9 reads DRV_EN to detect it.

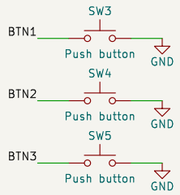
Hand-unit socket (case)



Hand unit (on the cable)



Analog joystick module (GND, VCC, VRx, VRy, SW), powered from 3.3 V.



Buttons and the joystick switch use the ESP32 internal pull-ups.

Notes

1. Supply, controller and drivers are all on the L: no cable crosses the azimuth joint.
2. +12V: 12.0–14.6 V from the battery or a lab supply. It feeds the drivers and U1.
3. TMC2209 header pin order differs between makers: wire by pin name. VIO at 3.3 V (ESP32 logic level).
4. Motor connectors: pins 1–2 one coil, 3–4 the other.
5. A1 pins: 1–22 = header J1, 23–44 = header J3.
6. Hand unit on an 8-wire cable (J2: for example GX16-8, or RJ45 with a network cable).
7. BZ1 directly on GPIO21: piezo buzzer, or active buzzer drawing at most 20 mA at 3.3 V.
8. Keep the blue LED dim at night.

All electronics on the L of the mount
ESP32-S3 controller, two TMC2209 drivers, 12.8 V LiFePO4 battery or lab supply

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Title: Star-Track mount electronics

Size: A3	Date: 2026-10-25	**Rev: 3**
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